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An alternative nonparametric tail risk measure

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The proposition of tail risk as a new asset pricing factor has gained traction in recent years. Recent work by Almeida, Ardison, Garcia, and Vicente (Nonparametric tail risk, stock returns, and the macroeconomy. *J. Financ. Economet.*, 2017, **15**(3), 333–376) proxies the cross-sectional variation in returns by Fama-French portfolios, which are further summarized into a few basis assets via principal component analysis. The number of states of nature is set higher than that of the basis assets to estimate the stochastic discount factors, which in turn risk-neutralize the excess expected shortfall as a tail risk measure. As an alternative approach to this means of dimension reduction, we propose tackling the problem directly by forming portfolios that minimize the excess expected shortfall. Our direct measure exhibits greater explanatory power when applied to more liquid, nonlottery-style stock returns. More importantly, our proposed approach reveals direct exposure to downside systematic risk without the need for an additional risk-neutralization step.

Keywords: Asset pricing; Risk neutralization; Risk optimization; Stochastic discount factor

1. Introduction

In modern portfolio theory, optimal asset allocation is performed by maximizing the reward-to-risk ratio in a mean-variance framework. As downside risk can weigh more heavily in the utility functions of risk-averse investors, mean-variance optimization can be an inefficient strategy because the traditional variance or standard deviation does not distinguish between positive and negative tails. Some researchers have attempted to resolve this issue by replacing variance with conditional value-at-risk, which addresses investors' asymmetric response to downside risk versus upside gain (see, e.g. Rockafellar and Uryasev 2000, Krokhmal *et al.* 2001).

The seminal attempt to formalize the pricing of downside risk in the cross-sectional variation in stock returns is that of Ang *et al*. (2006). In that paper, the authors work with a rational disappointment aversion utility function following the approach of Gul (1991). With the introduction of disappointment aversion to the utility function, which down-weights good outcomes relative to disappointing outcomes, the regular capital asset pricing model (CAPM) beta is no longer a sufficient statistic to describe the risk-return relationship of an individual stock. Through some measure of the relative downside beta, the optimal solution to the portfolio problem gives rise to a higher portfolio mean return and CAPM alpha with a higher downside beta, which justifies theoretically how downside risk can be a priced factor that is not captured by the CAPM. The empirical studies of Ang *et al*. (2006) also support the argument that stocks with higher downside betas have contemporaneously higher average returns that cannot be explained by the regular CAPM beta.

The foregoing approaches, while trying to treat downside risk as a priced factor, do not take the risk appetite of investors into consideration, i.e. the level of risk at which a given investor is willing to make a risky investment. Bali, Cakici *et al*. (2011) answer this question directly. They formalize a generalized measure of riskiness by defining it as the minimum wealth level at which a risky asset may be accepted. The minimum wealth level differs by the degree of risk aversion of the investor concerned. The authors further seek the unique real solution of the minimum wealth level under a constant relative risk-aversion utility function (Bali, Cakici *et al*. 2011). Based on this line of reasoning, the expected shortfall, defined as the expected loss once value-at-risk (VaR) has been breached, is a natural potential asset pricing factor, as it is a measure of the maximum loss (or minimum wealth) of an investment at a particular confidence level.

Recent works by Almeida and Garcia (2016) and Almeida *et al*. (2017) develop a nonparametric risk measure by risk-neutralizing the excess expected shortfall (the expected loss in excess of VaR in case of a VaR breach) and study it as an

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asset pricing factor. The main contribution of Almeida *et al.* (2017) is their use of the cross-sectional variation in returns to derive the stochastic discount factor (SDF) as opposed to such common option-based approaches as those of Aït-Sahalia and Lo (1998) and Aït-Sahalia and Lo (2000). When estimating the states of nature, Almeida *et al.* (2017) use a local window of the most recent 30 days to avoid pooling in old information. Because the number of states of nature needs to be higher than the number of basis assets under the usual assumption of an incomplete market, the authors perform a two-stage dimension reduction: they first use the Fama-French 25 size and book-to-market portfolios (Fama and French 1992) to approximate the cross-sectional variation in returns, and then conduct principal component analysis (PCA) to select the top five principal components. According to their empirical analysis, these five components represent around 90% of variation historically (Almeida *et al.* 2017). Their use of PCA is echoed by the recent work of Kozak *et al.* (2018), who report an SDF based on the first few principal components of asset returns to explain a number of anomalies. However, their choices of five principal components and a 30-day local window as the state of nature are not without controversy. Dobrev and Schaumburg (2017), for example, argue that the choices leave plenty of room for alternative implementations.

Motivated by the fact that the expected shortfall is a potential asset pricing factor that has not yet been fully explored in the literature, we posed the following question: Is it possible to construct a measure that resembles the best representation of tail risk from the cross-sectional variation in individual stocks directly? There were two key points to consider. First, we had to look for a construction of tail risk with an ultimate attempt to optimize downside risk, thus avoiding the need for further risk neutralization. Second, we wanted to avoid the prior construction of Fama-French portfolios for preliminary dimension reduction purposes.

We attack the problem by direct optimization. Instead of first replicating the universe of stocks by proxies, we directly minimize the excess expected shortfall. Our direct approach, hereafter denoted the minimum excess expected shortfall (MEES) method, allows us to estimate tail risk without first constructing Fama-French portfolios. By posing the problem as direct optimization with complementarity constraints, it can be efficiently solved by a standard algorithm such as sequential least squares programing. A nice interpretation of our direct construction is that the resulting MEES equates to the unavoidable or undiversifiable downside risk such that the compensation for exposure to it has a sound theoretical basis. With this in mind, we can also extend the traditional asset allocation decision by the mean-MEES frontier rather than existing mean-variance or mean-VaR metrics. Our empirical study also shows the MEES method to be superior to the approach proposed by Almeida *et al.* (2017) when applied to a more liquid universe; i.e. our result is less prone to size and liquidity effects. As a byproduct of MEES construction, the choice of the number of target portfolios was found to be critical in the evaluation of tail risk as an asset pricing factor. It is also a potential reflection of the downside synchronicity of cross-sectional security returns across a market.

The remainder of the paper is organized as follows. Section 2 describes the motivation underlying our MEES construction and the methodology used therein, and Section 3 outlines the characteristics of MEES. Section 4 applies MEES to three major markets, namely, the U.S., Japan, and Europe in a large-scale empirical study, and Section 5 concludes.

2. Motivation and methodology

Our research is an extension of the work in Almeida *et al.* (2017), who define tail risk (TR) as the risk-neutralized excess expected shortfall:

$$TR_{i,t} = E^{Q(R)}[(R_{i,j} - VaR_{\alpha}(R_{i,j}))|(R_{i,j} \leq VaR_{\alpha}(R_{i,j}))], \quad (1)$$

where t is the time period for which the TR is calculated, $j = 1, \ldots, J$ denotes the possible states of nature, α is the VaR threshold, and $Q(R)$ indicates the risk-neutral density. The aggregate market TR measure is the average of the $TR_{i,t}$ of several portfolios i.

The choice of J adopted by Almeida *et al.* (2017) is the 30 days prior to t, which corresponds to the state of nature, and α is set at 10%. In forming several portfolios i, Almeida *et al.* (2017) use the Fama-French 25 size and book-to-market portfolios, echoing the studies of Bali, Brown *et al.* (2014) and Kelly and Jiang (2014), and further extract via PCA the five that explain around 90% of the cross-sectional variation.

After forming the aggregate market TR, Almeida *et al.* (2017) run regressions for each stock in the Center for Research in Security Prices (CRSP) database with code (10)-(11) and measure the regression beta as TR hedging capacity. For each month from January 1967 to December 2013, the stocks are sorted and divided into 10 decile portfolios, and the returns of the portfolios are measured one-month post formation. According to Almeida *et al.* (2017), the average return of the lowest hedging portfolio (1st decile) exceeds that of the highest (10th decile) to a statistically significant degree. The αs of the regressions of the lowest hedging portfolio are also higher than those of the highest at a 5% level of significance under the Fama-French 3-factor model (Fama and French 1993). The results are similar using the 4-factor model, which includes the momentum factor (Carhart 1997), and the 5-factor model, which includes the liquidity factor (Pstor and Stambaugh 2003), suggesting that tail risk is an additional asset pricing factor that is not currently captured.

2.1. *Motivation for MEES construction*

In response to Almeida *et al.* (2017), who use a very wide universe of CRSP stocks to validate tail risk as a priced factor, Bali (2017) questions whether the outperformance they report could be the result of illiquid, value, momentum winners with a high degree of volatility and lottery-like characteristics. We investigate this possibility by replicating the Hellinger tail risk measure used by Almeida *et al.* (2017) but extracting it from S&P 500 index constituents instead of CRSP stocks.

Table 1. Replicating the Almeida *et al.* (2017) approach to S&P 500 stock returns.

Portfolio	Low	2	3	4	5	6	7	8	9	High	High - Low
Average return %	1.63	1.12	1.05	0.91	0.88	0.79	0.87	0.86	0.84	1.03	– 0.60
t-stat	2.72	2.55	2.45	2.45	2.53	2.35	2.73	2.55	2.25	1.97	– 1.26
FF3 alpha %	1.83	1.30	1.23	1.13	1.01	0.90	0.98	0.95	0.92	1.08	– 0.76
t-stat	2.85	3.10	2.78	3.18	2.92	2.72	3.17	2.88	2.60	2.20	– 1.45
FF5 alpha %	1.86	1.33	1.24	1.16	1.00	0.90	0.99	0.95	0.91	1.08	– 0.78
t-stat	2.84	3.08	2.76	3.18	2.86	2.68	3.15	2.87	2.59	2.27	– 1.48
FF5 + mom alpha%	1.85	1.33	1.24	1.15	1.00	0.89	0.98	0.95	0.91	1.08	– 0.78
t-stat	3.83	3.91	3.62	3.91	3.50	3.21	3.59	3.20	2.76	2.53	– 1.58

For each month from February 2002 to February 2018, the stocks are sorted based on the beta (using a 24-month rolling regression) with respect to the Hellinger tail risk, and the returns are tracked one-month post formation. The first line reports the average portfolio returns, with t-statistics given on the second line. The third, fifth, and seventh lines report the αs of regressions using the Fama-French 3-factor model, Fama-French 5-factor model, and Fama-French 5-factor model plus momentum factor, respectively, and the Newey-West (Newey and West 1987) t-statistics in the fourth, sixth, and eighth lines are all calculated with one lag.

All historical S&P 500 constituent data are sourced from Bloomberg.

The S&P 500 index constituents are the largest, most liquid stocks in the U.S., and we thus use this universe to address the question raised by Bali (2017). We perform PCA directly on the stock returns (as Fama-French portfolios are unavailable for S&P 500 consituents) over the period from February 2000 to February 2018. We find that for the 30 days prior to each month-end, the five principal components consistently capture 90% or more of the cross-sectional variations over the time period under study. We calculate the difference in returns between the highest and lowest hedging portfolios, as well as the alpha and its significance, under the FF 3-factor model (Fama and French 1993), FF 5-factor model (Fama and French 2015), and FF 5-factor model with the momentum factor (Carhart 1997), and find a considerable decline in statistical significance relative to the results of Almeida *et al.* (2017). See Table 1 for details.

The insignificant results presented in Table 1 indicate that the highly significant results in Almeida *et al.* (2017) may be due in part to the illiquidity or lottery-like characteristics of some of the CRSP stocks, particularly those in the lowest hedging portfolio. That possibility motivated us to design a more targeted, purer risk measure capable of reflecting the downside systematic risk such that the difference between the lowest and highest hedging portfolio is pronounced even in a liquid universe. To support our claim that we have indeed achieved a measure so capable, we apply the MEES measure to the S&P 500 index constituents in Section 4.1, and obtain a highly statistically significant difference in returns between the lowest and highest hedging portfolios.

2.2. MEES construction

Instead of representing a wide universe of securities by a few principal components, we construct a well-diversified portfolio from a universe that gives the MEES such that the beta exposure of each individual stock to this shortfall measure is the downside 'systematic' or unavoidable tail risk. As expected shortfall is a coherent and convex measure (see Artzner *et al.* 1999, Acerbi and Tasche 2002), a unique, global minimum solution is guaranteed. The problem is to solve for w_p^*:

$$w_p^* = \arg\min_{w_p} E(abs(w_p' R_t - VaR_\alpha(w_p' R_t)) \mid w_p' R_t \leq VaR_\alpha(w_p' R_t)), \tag{2}$$

subject to $w_p{}'1 = 1$, and $0 \leq w_p \leq 1$, where $R_t = (r_1, \ldots, r_n)'$ is the vector of returns for stocks 1 to n, α is the percentile for VaR, $1 = (1, \ldots, 1)'$ is simply a vector of ones with dimension $n \times 1$. We follow Almeida *et al.* (2017) in setting $\alpha = 10\%$.

However, as the market's downside systematic risk could be underestimated by forming only one portfolio, which may consist solely of stocks with a very low 'in-sample' excess shortfall (say, the 30 days before each month-end), we need to continue the process to extract additional portfolios to ensure that the measure properly represents the market-wide cross-sectional variation in stock returns. Ideally, the next portfolio p should have zero covariance with its predecessors. One might wonder why we do not reduce the universe by dropping the stocks we have already chosen and opt instead to extract further zero-covariance portfolios for wider exposure. Our weight analysis in Section 3.2 addresses this question by examining the weight composition of each portfolio and justifying our method of portfolio extraction.

Denoting the desired number of portfolios to be extracted from the universe by P, we look to solve further for $p = 2, \ldots, P$:

$$w_p^* = \arg\min_{w_p} E(abs(w_p' R_t - VaR_\alpha(w_p' R_t)) \mid w_p' R_t \leq VaR_\alpha(w_p' R_t)), \tag{3}$$

subject to $w_p{}'1 = 1$, $0 \leq w_p \leq 1$ and $cov(w_p' R_t, w_{p-1}' R_t) = \ldots = cov(w_p' R_t, w_1' R_t) = 0$.

We can then transform this problem into handy, direct optimization with complementarity constraints using samples for estimation. Adopting $J = 30$, i.e. the 30 days preceding each month-end t, as the states of nature, to allow direct comparison with the results in Almeida *et al.* (2017), we minimize the sample excess shortfall (of the portfolio) by optimizing the portfolio weights across the universe of stocks. More specifically, we solve for the weight vector $w_p = (w_{p,1}, \ldots w_{p,n})'$ of the p-th portfolio at each month-end t:

For $p = 1$,

$$\hat{w}_p = \arg\min_{w_p} \frac{1}{\alpha J} \sum_{j=1}^{J}_{w_p' R_{t-j} \le VaR_\alpha(w_p' R_t)} abs(w_p' R_{t-j} - VaR_\alpha(w_p' R_t)), \quad (4)$$

subject to $w_p' 1 = 1$, $0 \le w_p \le 1$. $VaR_\alpha(w_p' R_t)$ is VaR (at α-percentile) using the historical method of sample portfolio returns $w_p' R_{t-j}$ for $j = 1, \ldots, J$.

For $p = 2, \ldots, P$,

$$\hat{w}_p = \arg\min_{w_p} \frac{1}{\alpha J} \sum_{j=1}^{J}_{w_p' R_{t-j} \le VaR_\alpha(w_p' R_t)} abs(w_p' R_{t-j} - VaR_\alpha(w_p' R_t)),$$

subject to $w_p' 1 = 1$, and $0 \le w_p \le 1$ and $\sum_{j=1}^{J}(w_p' R_{t-j} - \overline{w_p' R})(w_{p-1}' R_{t-j} - \overline{w_{p-1}' R}) = \ldots = \sum_{j=1}^{J}(w_p' R_{t-j} - \overline{w_p' R})(w_1' R_{t-j} - \overline{w_1' R}) = 0$ and $\overline{w_p' R} = J^{-1} \sum_{j=1}^{J} w_p' R_{t-j}$.

The foregoing optimization with complementarity constraints can be solved efficiently by sequential least squares programing (see, e.g. Fletcher and Leyffer 2004). Standard tools such as SLSQP within the Python SciPy optimization package can also handle the problem efficiently.

With the desired number of target portfolios P, the MEES measure is defined simply as the average of the excess expected shortfalls of portfolios $p = 1, \ldots, P$:

$$MEES_t = \frac{1}{P} \sum_{p=1}^{P} \frac{1}{\alpha J} \sum_{\hat{w}_p' R_{t-j} \le VaR_\alpha(\hat{w}_p' R_t)} abs(\hat{w}_p' R_{t-j} - VaR_\alpha(\hat{w}_p' R_t)). \quad (5)$$

We cannot apply the same concept as Almeida *et al.* (2017), who use five portfolios extracted via PCA based on empirical 90% explanation of cross-sectional variation, as we are forming minimized excess expected shortfall portfolios instead of performing spectral decomposition. The higher the p-th portfolio we extract, the higher the resultant risk because the p-th portfolio is constructed to have zero covariance with the preceding portfolios. The empirical study in Section 4.6 will shed further light on our selection of P.

3. Characteristics of MEES

3.1. Evolution of MEES over time

Using the S&P 500 as the universe for our calculation of MEES, we plot the excess expected shortfall for extracted portfolios $p = 1, p = 3$, and $p = 5$ in Figure 1. A general observation to be noted is that the excess expected shortfall becomes spikier with a higher p. This is natural, as the highest p-th portfolio is constructed to have zero covariance with its predecessors such that the weights are forced into higher-risk stocks.

For example, when we look at the initial portfolio ($p = 1$), the excess expected shortfall exhibits no 'spikes' even in such typical high-risk periods as the Lehman Brothers collapse in October 2008, the start of the European sovereign debt crisis in 2010, and the S&P downgrade of U.S. long-term credit in late 2011. Such insensitivity may be the result of the portfolio's heavy tilt toward stocks with very low 'in-sample' excess shortfalls, meaning that the market-wide risk is under-represented. Thus, the process of extracting higher p portfolios with zero covariance with previous portfolios is crucial to ensuring that the MEES measure covers a sufficiently wide spectrum of stocks to accurately represent the market-wide downside risk. When $p = 3$, we see more spikes in the aforementioned high-risk periods. However, as p increases further to a relatively high level such as $p = 5$, we can observe the excess shortfall becoming very spiky, which might pool in too much idiosyncratic risk. This phenomenon motivates our empirical search for an optimal P in Section 4.6.

3.2. MEES weight analysis and compositions across sectors

In an attempt to justify why we extract further portfolios instead of reducing constituents in our construction of the MEES measure, Table 2 reports the average proportion of stocks with weights greater than zero for each portfolio $p = 1, 2, 3, 4, 5$ over the period from February 2000 to February 2018. In the first line of the table, we can see that over 90% of the constituent stocks have non-zero weights on average for all of the portfolios. The second line reports the weights of the heaviest constituents in each portfolio at each month-end averaged over the period. It can be seen that the average weight of the heaviest constituent over the period is increasing with p. This observation is in line with our intuition, as the portfolio with the highest p is constructed to have zero covariance with those with lower ps, and it is thus inevitable that more idiosyncratic risk is extracted, i.e. more corner solutions to maintain zero covariance with the preceding portfolios.

Figure 2 shows the weights of stocks belonging to each sector under the GICS level I† classification. We can see that the weights are fairly evenly spread across the sectors for all portfolios. Although the average maximum weight of portfolio $p = 1$ is low (Table 2), that portfolio's sector weights seem to be more widely dispersed than those of the other portfolios. It has the highest concentration in six of the 11 sectors and the lowest in four relative to the portfolios with a higher p. For the portfolio with the highest p, although the average weight of its heaviest constituent is greater than that of the other portfolios, its average sector weights are better aligned. This is not a

† GICS stands for Global Industry Classification Standard, an industry taxonomy developed by MSCI and Standard & Poor's. It is the benchmark for industry classification. GICS consists of 11 sectors (Level I), 24 industry groups (Level II), 68 industries (Level III), and 157 sub-industries (Level IV).

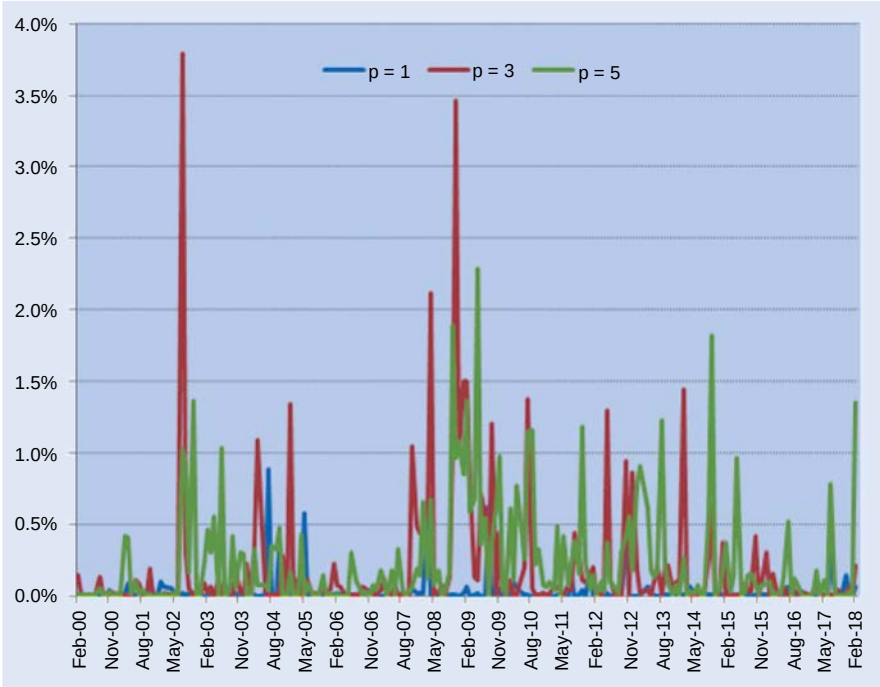

Figure 1. Evolution of minimum shortfalls for portfolios $\boldsymbol{p} = 1$, $\boldsymbol{p} = 3$, and $\boldsymbol{p} = 5$ from February 2000 to February 2018.

Table 2. Weight analysis of MEES portfolios for $\mathbf{p} = 1, 2, 3, 4, 5$ (S&P 500 universe).

	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$
proportion of stockswith weights > 0	99.90%	94.24%	92.72%	92.79%	94.02%
average maximum weight over the period	1.46%	12.52%	28.78%	39.17%	43.83%

Over the February 2000 to February 2018 period, the constituent stocks with weights greater than zero are counted for each month-end and divided by the total number of constituents to calculate the proportion for each period. The first line reports the proportions averaged over the entire period, and the second the weights of the heaviest stocks in each portfolio averaged over the period.

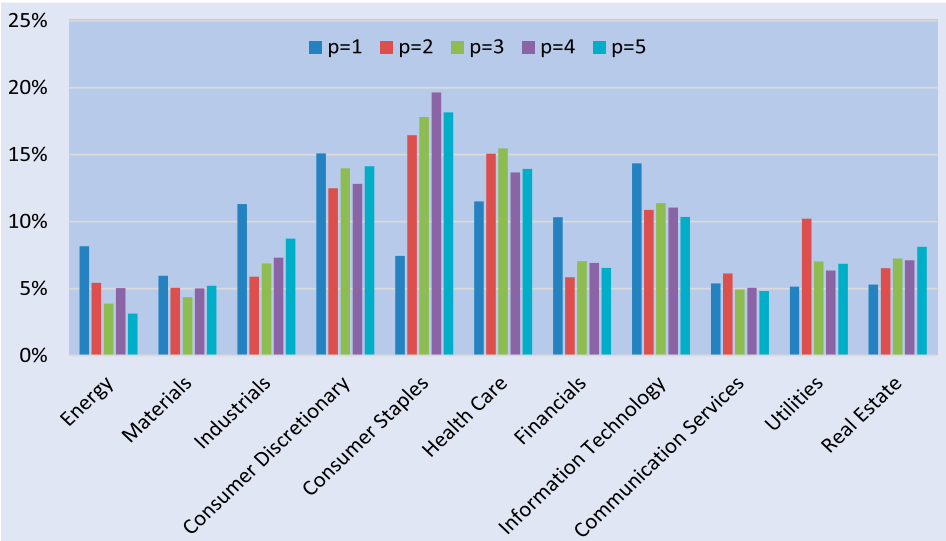

Figure 2. Sector allocation of portfolios $\boldsymbol{p} = 1, 2, 3, 4, 5$ (S&P 500 universe) over the February 2000 to February 2018 period.

contradictory finding, as the heaviest constituents at different month-ends can be in different sectors. Table 2 and Figure 2 report the averages over the period as a summary.

The stock weights within each sector are summed for each month and then averaged over the February 2000 to February 2018 period. The bars are the average weights of each sector for portfolios $p = 1, 2, 3, 4, 5$.

As we can see from Table 2 and Figure 2, the number of stocks with a non-zero weight is quite high, and the weights are evenly spread across the 11 industry sectors. Accordingly,

Table 3. Correlation matrix of MEES, Hellinger, and VIX measures from February 2000 to February 2018.

	MEES	Hellinger	VIX
MEES			
Hellinger	23.1%		
VIX	54.1%	38.6%	

we opted to extract additional portfolios rather than omit constituent stocks to obtain a better representation of the market-wide tail risk during our construction of the MEES measure.

3.3. *Comparison with other risk measures*

In this section, we compare the MEES measure (with $P = 5$) to the Hellinger risk measure (Almeida *et al*. 2017) using the S&P 500 as the universe for calculation. We also compare both measures with the S&P 500 volatility index (VIX). Table 3 is a correlation matrix of the three measures based on monthly data from February 2000 to February 2018. The moderate correlation between the MEES and Hellinger measures probably explains why the former works better than the latter with a more liquid universe such as the S&P 500 (see the results in Section 4.1).

Figure 3 plots the 24-month rolling correlations of MEES vs. Hellinger and MEES vs. VIX. We can see that MEES behaves very differently from the other two measures in certain periods, such as late-2004–2008 and 2013 to mid-2017. The non-synchronized correlations indicate that MEES is a stand-alone risk measure rather than a duplication of existing measures.

4. Empirical study

4.1. *Does risk neutrality matter? The case of S&P 500*

The idea of risk neutrality is to find an SDF that places more weight on relatively poor states of nature and less on relatively good ones. Almeida *et al*. (2017) found their risk neutrality exercise to significantly boost the statistical significance of the difference in returns between the highest and lowest hedging portfolios they considered. Our conjecture was that if we were to construct the excess expected shortfall by directly optimizing the downside systematic risk, then there would be no need for further risk neutralization because risk exposure would already have been dealt with directly during said construction. Here, we use the S&P 500 as the universe to determine whether the proposed MEES measure can show significance for the difference in returns between the highest and lowest hedging portfolios, for which insignificance was shown using the Hellinger tail risk measure in Table 1.

Table 4 reports the difference in returns between the highest and lowest hedging portfolios using the MEES measure, as well as the alpha and its significance, under the FF 3-factor model (Fama and French 1993), FF 5-factor model (Fama and French 2015), and FF 5-factor model with the momentum factor (Carhart 1997). Two interesting observations can be made. First, the negative t-statistics become increasingly significant as P increases up to 4. This pattern could be the result of the MEES measure not capturing the market-wide risk when P is low. The t-statistics jump when P increases to 2 and then to 3, eventually reaching an optimum when P equals 4. The degree of significance drops slightly when P increases to 5, which may imply that an excessive degree of risk is being extracted from the stock universe. This finding motivates our optimal selection of Pin Section 4.6. Second, applying the same risk-neutrality procedure as that in Almeida *et al*. (2017) in our optimization, which corresponds to the case of 'Risk neutral = True' in Table 4, seems to make no difference when compared to the case of no risk-neutral adjustment (see the right-hand side of Table 4). In fact, the results seem to argue against risk neutralization except in the case of $P = 5$, which confirms our conjecture that when MEES is directly constructed to minimize the excess expected shortfall, there is no need for further risk neutralization.

4.2. *Beta persistence: tail risk as a priced factor*

For tail risk to be a valid asset pricing factor, it needs to be cross-sectionally peristent over time. To show this, we track

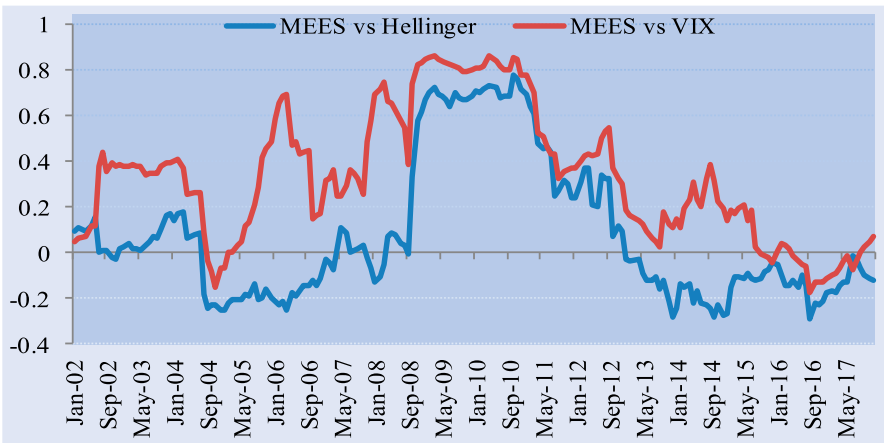

Figure 3. Rolling 24-month correlation of MEES vs. Hellinger measure and MEES vs. VIX.

Table 4. Difference between the average returns of the highest and lowest hedging portfolios to MEES for the S&P 500 constituents from February 2002 to February 2018.

	Risk neutral = True					Risk neutral = False				
S&*P* 500	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$
Average return	– 0.13	– 0.42	– 1.04	– 1.22	– 1.20	– 0.14	– 0.56	– 1.26	– 1.39	– 1.16
t-statistics	– 0.36	– 1.50	**– 2.60**	**– 3.20**	**– 3.06**	– 0.39	**– 2.05**	**– 3.07**	**– 3.38**	**– 2.49**
FF3 alpha	– 0.19	– 0.50	– 1.07	– 1.22	– 1.20	– 0.20	– 0.60	– 1.27	– 1.37	– 1.15
t-statistics	– 0.55	**– 1.70**	**– 2.43**	**– 2.89**	**– 2.70**	– 0.56	**– 2.26**	**– 2.83**	**– 2.97**	**– 2.22**
FF5 alpha	– 0.18	– 0.50	– 1.07	– 1.24	– 1.38	– 0.19	– 0.60	– 1.28	– 1.38	– 1.14
t-statistics	– 0.53	**– 1.70**	**– 2.41**	**– 2.88**	**– 2.94**	– 0.55	**– 2.27**	**– 2.80**	**– 2.94**	**– 2.14**
FF5 + mom alpha	– 0.18	– 0.50	– 1.07	– 1.24	– 1.37	– 0.19	– 0.60	– 1.28	– 1.37	– 1.16
t-statistics	– 0.60	**– 1.79**	**– 3.08**	**– 3.53**	**– 3.87**	– 0.62	**– 2.28**	**– 3.60**	**– 3.87**	**– 2.58**
beta persistence t-statistics	12.08	17.72	19.23	13.97	17.81	12.24	18.70	19.92	13.91	13.01
proportion of low hedging at t + 1	0.95	0.96	0.96	0.96	0.97	0.95	0.96	0.96	0.96	0.96
proportion of high hedging t + 1	0.98	0.97	0.97	0.97	0.97	0.98	0.97	0.97	0.97	0.96

For each month from February 2002 to February 2018, the stocks are sorted based on the beta (based on a rolling regression of 24 months) with respect to MEES and the returns are tracked one-month post formation. The first line reports the average portfolio return difference (%) between the highest and lowest hedging portfolios, with the t-statistics in the second line for MEES constructed with $P = 1, 2, 3, 4, 5$. The third line reports the αs of regressions (%) based on the Fama-French 3-factor model, with t-statistics in the fourth line. The fifth and seventh lines report the αs of regressions using the Fama-French 5-factor model and Fama-French 5-factor model with the momentum factor, respectively, and the sixth and eighth lines report the Newey-West (Newey and West 1987) t-statistics with one lag.

the constituents of the highest and lowest hedging portfolios at each time t and measure their average betas at time $t+1$ in the cases of $P = 1, 2, 3, 4, 5$. If tail risk persists, then the difference between the two average betas (over the time period) should be statistically significant (see 'beta persistence t-statistics' in the lower part of Table 4). We can see from the table that the t-statistics are significant in all cases ($P = 1, 2, 3, 4, 5$).

We also follow the more rigorous approach of Bali, Cakici, and Whitelaw (2011) in studying the beta persistence. Here, we examine the proportion of stocks in the lowest hedging portfolio at time t, and find that the majority of stocks remain in the three lowest deciles at time $t+1$, as reported in the line following 'proportion of low hedging at $t+1$' in the lower part of Table 4. Similarly, the table shows that the greatest proportion of stocks in the highest hedging portfolio at time $t$still appear in the three highest deciles at time $t+1$ ('proportion of high hedging t + 1'). All of the proportions are 95% or above, indicating the substantial cross-sectional persistence of tail risk.

The lower portion of the table reports the persistence of tail risk. The first line reports the t-statistics of the difference in the average beta of the highest hedging portfolio (sorted at time t) and that of the lowest hedging portfolio (sorted at time t) at time t + 1. The second line shows that the average proportion of constituents in the lowest hedging portfolio (sorted at time t) still exists in the three lowest deciles at time t + 1, and the third that the average proportion of constituents in the highest hedging portfolio (sorted at time t) still exists in the three highest deciles at time t + 1.

4.3. *Predictive properties of MEES measure*

To further compare the performance of our MEES measure (without risk neutrality adjustment) with the Hellinger measure and VIX, we study the predictive power of each for future returns by the following regression.

$$r_{[t,t+k]} = \alpha + \beta TR_t + \gamma r_{[t-k,t]} + u_t, \tag{6}$$

where TR_t denotes the tail risk measure. We take seven cases into consideration: the MEES measure at $P = 1, 2, 3, 4, 5$, the Hellinger measure extracted from the S&P 500, and VIX. We use S&P 500 returns as $r_{[t,t+k]}$ to be predicted for $k = 1, \ldots, 12$ months. Because our dataset spans the months from February 2000 to February 2018, the periods for the predictive regressions are shortened at the beginning and end by k lag in each case of k.

Table 5 reports the results of the predictive regressions. It can be seen that the MEES measure works well for $P = 2, 3, 4, 5$ in the four- to seven-month horizons, whereas the Hellinger measure (extracted from the S&P 500) is not a significant predictor of future returns in any of the horizons. The signs of the betas even point in an undesirable direction. Although the beta of VIX exhibits the right sign, there is a notable drop in its significance when we compare it with the time horizon studied by Almeida *et al.* (2017), that is, January 1991 to April 2014. The implication is that VIX's predictive power for future returns declined between April 2014 and February 2018.

4.4. *Applying MEES extracted from the S&P 500 to CRSP stocks*

To determine how the MEES measure performs if extracted from a small liquid universe but applied to a broader universe, we extend our comparison of the Hellinger risk measure and MEES measure in Tables 6 and 7. Both measures are extracted from the S&P 500 but applied to the broader universe of CRSP stocks. In other words, we replicate Tables 1 and 4 in Tables 6 and 7, respectively, but the portfolios are sorted with CRSP stocks. As can be seen from the two tables,

Table 5. Predictive regression beta of MEES measure, Hellinger measure, and VIX from February 2000 to February 2018.

Beta	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	Hellinger	VIX
k = 1	2.26	3.05	0.49	0.07	0.13	– 0.22	0.00
2	3.31	6.32	– 0.41	0.63	1.19	– 0.30	0.01
3	2.80	6.50	2.67	2.69	4.18	– 0.42	0.11
4	4.42	10.04	6.83	6.62	8.44	– 0.61	0.14
5	3.53	11.61	9.41	8.50	11.38	– 0.44	0.19
6	2.41	14.33	11.45	9.63	13.02	– 0.36	0.22
7	0.82	13.55	10.53	9.81	13.55	– 0.31	0.22
8	– 2.25	13.42	11.12	10.78	14.44	– 0.24	0.18
9	– 4.82	11.46	13.36	11.73	16.13	– 0.34	0.20
10	– 8.22	6.17	12.46	11.65	16.71	– 0.48	0.24
11	– 8.48	8.44	13.04	11.81	17.04	– 0.47	0.28
k = 12	– 7.65	9.42	15.35	13.43	18.48	– 0.41	0.28
t-stat	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	Hellinger	VIX
k = 1	1.48	1.22	0.31	0.05	0.08	– 0.54	0.05
2	1.47	**1.86**	– 0.13	0.25	0.43	– 0.45	0.12
3	1.27	**1.86**	0.82	1.03	1.48	– 0.54	0.87
4	**1.72**	**2.15**	**1.93**	**2.24**	**2.56**	– 0.59	0.91
5	0.97	**2.40**	**2.20**	**2.54**	**3.08**	– 0.41	1.29
6	0.67	**2.75**	**2.08**	**2.15**	**2.73**	– 0.32	1.44
7	0.12	**2.32**	**1.70**	**1.94**	**2.46**	– 0.22	1.27
8	– 0.27	**1.65**	1.50	**1.82**	**2.28**	– 0.14	0.90
9	– 0.61	1.26	**1.74**	**1.81**	**2.28**	– 0.17	0.88
10	– 0.85	0.63	1.44	1.49	**1.95**	– 0.23	0.87
11	– 0.76	0.78	1.49	1.42	**1.84**	– 0.21	0.88
k = 12	– 0.70	0.84	**1.66**	1.49	**1.90**	– 0.17	0.77

The returns of the S&*P* 500 index are predicted using the regression $r_{[t,t+k]} = \alpha + \beta TR_t + \gamma r_{[t-k,t]} + u_t$ over the period from February 2000 to February 2018. Here, TR_t is the tail risk measure. We take seven cases into consideration: the MEES for $P = 1, 2, 3, 4, 5$, the Hellinger measure, and VIX. The t-statistics are calculated using Newey-West (Newey and West 1987) variance with 2–24 lags for 1–12 forecasting horizons.

Table 6. Replicating Table 1 with Hellinger measure extracted from the S&*P* 500 but applied to CRSP stocks from February 2002 to February 2018.

Portfolio	Low	2	3	4	5	6	7	8	9	High	High - Low
Average return %	12.44	2.49	2.07	3.97	2.47	2.72	2.14	2.82	3.85	6.19	– 6.25
t-stat	2.43	3.77	4.09	4.32	3.28	2.82	2.86	3.17	2.40	4.76	– 1.18
FF3 alpha %	13.71	2.62	2.18	4.03	2.45	3.28	2.36	3.08	3.94	6.46	– 7.26
t-stat	2.28	4.09	4.30	4.30	3.45	2.50	3.08	3.06	2.29	5.05	– 1.18
FF5 alpha %	13.69	2.60	2.21	3.93	2.45	3.32	2.35	3.03	4.05	6.53	– 7.15
t-stat	2.28	4.04	4.31	4.45	3.35	2.45	3.04	3.00	2.21	4.94	– 1.16
FF5 + mom alpha%	13.68	2.60	2.20	3.93	2.44	3.32	2.35	3.03	4.04	6.53	– 7.15
t-stat	2.27	4.54	4.87	4.53	3.45	2.45	3.12	3.03	2.20	4.94	– 1.16

For each month from February 2002 to February 2018, the CRSP stocks are sorted based on the beta (using a 24-month rolling regression) with respect to the Hellinger tail risk (extracted from the S&*P* 500), and the returns are tracked one-month post formation. The other notations are the same as in Table 1.

although the degree of statistical significance declines in both cases, the MEES measure still performs noticeably better than the Hellinger measure.

4.5. *Other markets: the case of Japan and Europe*

We next apply the MEES measure to the second largest stock market in the world, namely, Japan, using the broad, liquid TPX 500 as the universe for MEES construction. Using the same procedure as that in Table 4 to calculate the beta exposure of each stock with respect to MEES and sort the stocks into 10 decile portfolios, we obtain the results reported in Table 8 for the difference between the average returns of the highest and lowest hedging portfolios.

Table 8 underscores the two important facts we saw in the case of the S&P 500. First, the t-statistics for MEES with a lower *P* exhibit undesirable positive signs, i.e. the lowest hedging portfolio has a lower return than the highest. The t-statistics turn negative only when *P*increases to 3, and become significant (at a 5% significance level) when itincreases to 7. Second, the t-statistics again document evidence of the non-necessity of risk-neutral adjustment. The t-statistics of the beta persistence are significant for all lags, and the constituents of the lowest and highest hedging portfolios (sorted

Table 7. Replicating Table 4—Difference between the average returns of the highest and lowest hedging portfolios to MEES (extracted from the S&*P* 500) for CRSP constituents from February 2002 to February 2018.

	Risk neutral = True					Risk neutral = False				
	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$
Average return	0.14	– 4.89	– 9.69	– 9.05	– 9.29	– 0.70	– 5.82	– 9.97	– 8.31	– 8.96
t-statistics	0.18	– 1.09	**– 1.91**	**– 1.78**	**– 1.84**	– 0.62	– 1.30	**– 1.97**	**– 1.66**	**– 1.79**
FF3 alpha	0.02	– 5.76	– 10.66	– 10.12	– 10.33	– 0.72	– 6.72	– 10.95	– 9.48	– 10.04
t-statistics	0.02	– 1.05	**– 1.80**	**– 1.71**	**– 1.74**	– 0.68	– 1.22	**– 1.85**	– 1.61	**– 1.70**
FF5 alpha	0.02	– 5.85	– 10.76	– 10.21	– 9.49	– 0.75	– 6.83	– 11.06	– 9.49	– 10.05
t-statistics	0.03	– 1.06	**– 1.81**	**– 1.72**	– 1.61	– 0.68	– 1.23	**– 1.86**	– 1.61	**– 1.70**
FF5 + mom alpha	0.02	– 5.85	– 10.76	– 10.22	– 9.49	– 0.75	– 6.83	– 11.06	– 9.49	– 10.06
t-statistics	0.03	– 1.05	**– 1.80**	**– 1.71**	– 1.61	– 0.68	– 1.23	**– 1.86**	– 1.61	**– 1.70**
beta persistence t-statistics	11.42	18.26	13.73	11.91	16.68	11.56	15.61	13.81	12.86	10.43
proportion low hedging t + 1	0.95	0.96	0.96	0.96	0.96	0.95	0.96	0.96	0.96	0.95
proportion high hedging t + 1	0.96	0.96	0.99	0.99	0.99	0.96	0.96	0.98	0.99	0.99

For each month from February 2000 to February 2018, the CRSP stocks are sorted based on the beta (based on a rolling regression of 24 months) with respect to MEES (extracted from the S&*P* 500), and the returns are tracked one-month post formation. The other notations are the same as those in Table 4.

Table 8. Difference between the average returns of the highest and lowest hedging portfolios to MEES for TPX 500 constituents from March 2007 to June 2018.

	Risk neutral = True					Risk neutral = False				
TPX 500	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$
Average return	0.44	0.61	– 0.37	– 0.59	– 0.67	0.48	0.35	– 0.26	– 0.60	– 0.64
t-statistics	1.56	1.47	– 0.70	– 1.25	– 1.39	1.71	0.83	– 0.50	– 1.25	– 1.42
FF3 alpha	0.23	0.23	– 0.76	– 0.66	– 0.75	0.26	– 0.06	– 0.69	– 0.72	– 0.75
t-statistics	0.75	0.54	– 1.28	– 1.19	– 1.30	0.85	– 0.14	– 1.21	– 1.38	– 1.47
FF5 alpha	0.31	0.25	– 0.87	– 0.78	– 0.89	0.33	– 0.04	– 0.79	– 0.79	– 0.87
t-statistics	1.00	0.60	– 1.42	– 1.39	– 1.54	1.06	– 0.10	– 1.37	– 1.52	**– 1.71**
FF5 + mom alpha	0.32	0.28	– 0.82	– 0.74	– 0.85	0.34	– 0.01	– 0.75	– 0.75	– 0.84
t-statistics	1.04	0.73	**– 1.65**	– 1.59	**– 1.78**	1.10	– 0.03	– 1.61	**– 1.71**	**– 1.99**
beta persistence t-statistics	11.45	10.59	18.73	24.37	26.91	11.75	12.42	18.68	25.14	25.94
Prop low hedging t + 1	0.96	0.96	0.97	0.97	0.97	0.96	0.97	0.97	0.97	0.97
Prop high hedging t + 1	0.98	0.97	0.97	0.97	0.98	0.98	0.98	0.97	0.97	0.97
	$P=6$	$P=7$	$P=8$	$P=9$	$P=10$	$P=6$	$P=7$	$P=8$	$P=9$	$P=10$
Average return	– 0.72	– 0.70	– 0.68	– 0.66	– 0.65	– 0.62	– 0.79	– 0.85	– 0.85	– 0.88
t-statistics	– 1.58	– 1.54	– 1.55	– 1.52	– 1.46	– 1.40	**– 1.80**	**– 1.92**	**– 1.95**	**– 2.03**
FF3 alpha	– 0.72	– 0.78	– 0.74	– 0.73	– 0.78	– 0.70	– 0.87	– 0.93	– 0.96	– 0.97
t-statistics	– 1.39	– 1.46	– 1.42	– 1.43	– 1.47	– 1.40	**– 1.76**	**– 1.78**	**– 1.86**	**– 1.92**
FF5 alpha	– 0.86	– 0.91	– 0.91	– 0.88	– 0.91	– 0.83	– 0.99	– 1.04	– 1.06	– 1.08
t-statistics	**– 1.69**	**– 1.75**	**– 1.76**	**– 1.74**	**– 1.75**	**– 1.66**	**– 2.03**	**– 2.01**	**– 2.10**	**– 2.15**
FF5 + mom alpha	– 0.82	– 0.88	– 0.87	– 0.85	– 0.88	– 0.79	– 0.96	– 1.00	– 1.03	– 1.05
t-statistics	**– 1.98**	**– 2.05**	**– 2.02**	**– 1.95**	**– 1.97**	**– 1.89**	**– 2.30**	**– 2.31**	**– 2.41**	**– 2.44**
beta persistence t-statistics	25.14	23.17	22.64	21.81	20.86	25.59	23.97	24.52	23.38	22.90
Prop low hedging t + 1	0.97	0.97	0.96	0.96	0.96	0.97	0.96	0.96	0.96	0.96
Prop high hedging t + 1	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97

For each month from March 2007 to June 2018, the stocks are sorted based on the beta (based on a rolling regression of 24 months) with respect to MEES, and the returns are tracked one-month post formation. The first line reports the average portfolio return difference (%) between the highest and lowest hedging portfolios, with the t-statistics in the second line for MEES constructed with $P = 1, \ldots, 10$. The other notations are the same as those in Table 4.

at time t) remain primarily in the three lowest and three highest deciles at time $t + 1$.

We now extend the MEES measure to Europe. We use the Euro Stoxx index (SXXE), which consists of the approximately 300 most liquid stocks from 12 Eurozone countries. Applying the same procedures as we did in the cases of the S&P 500 and TPX 500, we find the most significant t-statistics for SXXE to occur at $P = 1$ (see Table 9).

4.6. The choice of P: target number of portfolios to be extracted

We have seen that the target number of portfolios to be extracted, *P*, plays an important role in the significance of the difference in returns between the highest and lowest hedging portfolios. Moreover, the optimal *P* seems to be market-specific.

Table 9. Difference between the average returns of the highest and lowest hedging portfolios to MEES for SXXE from March 2004 to June 2018.

	Risk neutral = True					Risk neutral = False				
SXXE	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$	$P=1$	$P=2$	$P=3$	$P=4$	$P=5$
Average return	– 0.82	– 0.15	– 0.49	– 0.30	– 0.15	– 0.75	– 0.19	– 0.24	– 0.38	– 0.11
t-statistics	**– 2.23**	– 0.42	– 1.12	– 0.75	– 0.35	**– 2.04**	– 0.47	– 0.59	– 0.94	– 0.26
FF3 alpha	– 1.20	– 0.24	– 0.50	– 0.37	0.02	– 1.14	– 0.23	– 0.28	– 0.43	0.03
t-statistics	**– 2.45**	– 0.63	– 0.94	– 0.78	0.05	**– 2.36**	– 0.51	– 0.58	– 0.86	0.07
FF5 alpha	– 1.11	– 0.14	– 0.38	– 0.29	0.06	– 1.04	– 0.10	– 0.17	– 0.39	0.08
t-statistics	**– 2.47**	– 0.39	– 0.77	– 0.65	0.12	**– 2.36**	– 0.24	– 0.38	– 0.83	0.17
FF5 + mom alpha	– 1.53	– 0.60	– 1.04	– 0.88	– 0.65	– 1.44	– 0.65	– 0.78	– 1.00	– 0.62
t-statistics	**– 3.49**	– 1.61	**– 2.49**	**– 2.22**	– 1.62	**– 3.30**	**– 1.70**	**– 1.96**	**– 2.47**	– 1.47
beta persistence t-statistics	8.18	21.84	19.08	21.52	18.58	8.54	23.19	21.28	18.17	15.47
proportion of low hedging t + 1	0.95	0.96	0.96	0.97	0.97	0.95	0.96	0.96	0.97	0.97
proportion of high hedging t + 1	0.97	0.98	0.97	0.98	0.98	0.97	0.98	0.97	0.97	0.98

For each month from March 2004 to June 2018, the SXXE stocks are sorted based on the beta (based on a rolling regression of 24 months) with respect to MEES, and the returns are tracked one-month post formation. The first line reports the average portfolio return difference (%) between the highest and lowest hedging portfolios, with the t-statistics in the second line for MEES constructed with $P = 1, 2, 3, 4, 5$. The other notations are the same as those in Table 4.

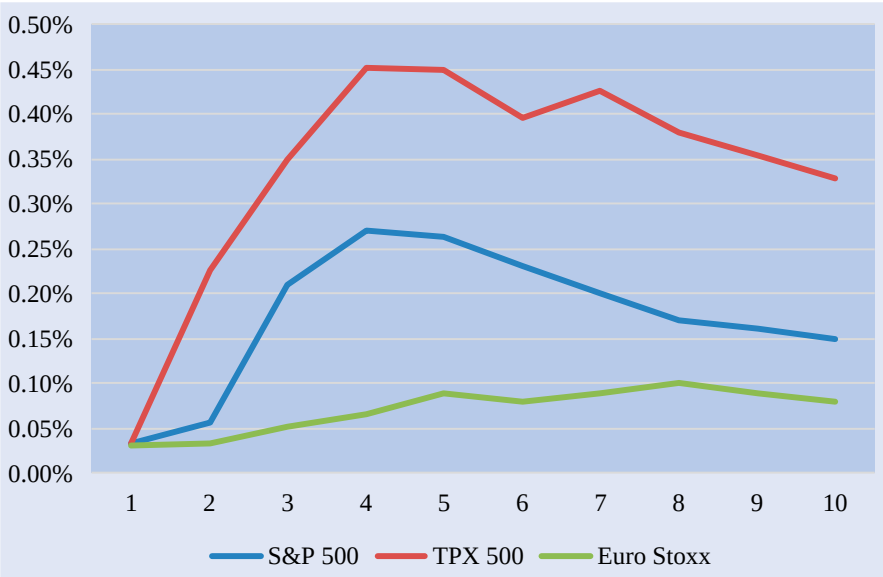

Figure 4. Average monthly MEES for S&P 500, TPX 500, and SXXE for $\boldsymbol{p} = 1, \ldots, 10$.

Because the excess expected shortfall is the difference between the expected shortfall and VaR, it is likely be high if the market's degree of downside synchronicity is low, i.e. there are some stocks in the tail that behave very poorly and some that are just marginally below VaR. In markets with a high degree of downside synchronicity, in contrast, VaR will also be high, and hence the excess expected shortfall low. As we want a wide representation of the market tail risk, P needs to be higher for markets in which the downside returns are highly dispersed (as in the case of the TPX 500), and the excess expected shortfall should plateau or steadily drop after that P.

Figure 4 plots the average monthly MEES (over our study periods) without risk-neutral adjustment for the three markets considered with different Ps. The results are similar for risk-neutral cases, which are not shown here. We can immediately notice one interesting point: the TPX 500 has a much higher MEES magnitude than the other markets, which could mean that its downside synchronicity of returns is much more dispersed, thus resulting in a higher excess expected shortfall. The point after which we see a steady decrease in that shortfall is $P = 7$ (see Table 8), which could mean that the downside dispersion begins to diminish because VaR adequately represents the risk level. Therefore, for MEES at $P = 7$, we start to see statistical significance for the difference in returns between the highest and lowest hedging portfolios. When similar reasoning is applied to the case of the S&P 500, we obtain $P = 4$ (see Table 4).

We can see from the figure that the excess expected shortfall is much lower in magnitude for the SXXE than it is for the two other markets, which could imply a generally high downside synchronicity of returns across the Eurozone; i.e. the VaR level is close to the expected shortfall. We can also see that $P = 1$ produces the greatest statistical discrepancy between the returns of the highest and lowest hedging portfolios, which seems consistent with our intuition that the entire

Eurozone has been beleaguered by a certain degree of market default risk or system breakdown during our period of study. However, we expected statistical significance to persist beyond $P = 1$, and hence the noticeable drop in the t-statistics from $P = 2$ onwards in Table 9 is puzzling.

5. Conclusions

In this research, we present a new direct approach to tail risk construction, dubbed MEES, that extracts minimum excess expected shortfall portfolios with zero between-portfolio covariance. Such direct construction avoids the need to construct Fama-French portfolios in the first instance, meaning that our approach can be readily applied to any markets in which accounting standards do not allow the construction of accurate Fama-French-type portfolios. Also, by posing our problem as direct optimization with complementarity constraints, we ensure that it can be efficiently solved by a standard algorithm such as sequential least squares programing. In addition, our 'minimum' excess expected shortfall construction offers a nice interpretation, namely, that the MEES measure equates to the unavoidable or undiversifiable downside risk, and thus that the return compensating for risk exposure has a sound theoretical basis. The empirical study in Section 4 not only shows MEES to outperform the Hellinger tail risk measure when applied to a smaller but more liquid stock universe, i.e. our result is less prone to capturing returns due to the lottery-like characteristics of small, illiquid stocks, but also sheds light on the target number of portfolios to be extracted, thereby affording us a sense of the synchronicity of the downside returns of securities across a given market.

A fruitful direction for further research would be to consider maximum rather than minimum risk construction, i.e. a tail risk measure that maximizes the potential excess expected shortfall. Such a measure might be able to capture tail risk with fewer portfolios. However, as the expected shortfall is a convex function, bounds might need to be imposed on the weight vector (domain) for the problem to have a solution, and it might be a corner solution, thus adding further complexity. The proper identification and setup of the maximization problem would be an interesting direction for tail risk construction.

Finally, it should be noted that there are a number of other tail risk measures, including those proposed by Bloom (2009), Kelly and Jiang (2014), Allen *et al*. (2012), and Bollerslev *et al*. (2015). Because our work constitutes an alternative approach to the non-parametric construction in Almeida *et al*. (2017), we do not compare it directly with other parametric or option-based measures. A direction for future research would be to conduct a detailed comparison of the tail risks obtained using different approaches.

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References

Acerbi, C. and Tasche, D., Expected shortfall: A natural coherent alternative to value at risk. *Econom. Notes*, 2002, **31**(2), 379–388.
Aït-Sahalia, Y. and Lo, A. W., Nonparametric estimation of state-price densities implicit in financial asset prices. *J. Financ.*, 1998, **53**(2), 499–547.
Aït-Sahalia, Y. and Lo, A. W., Nonparametric risk management and implied risk aversion. *J. Econom.*, 2000, **94**(1), 9–51.
Allen, L., Bali, T. G. and Tang, Y., Does systematic risk in the financial sector predict future economic downturns? *Rev. Financ. Stud.*, 2012, **25**, 3000–3036.
Almeida, C. and Garcia, R., Economic implications of nonlinear pricing kernels. *Manage. Sci.*, 2016, **63**(10), 3361–3380.
Almeida, C., Ardison, K., Garcia, R. and Vicente, J., Nonparametric tail risk, stock returns, and the macroeconomy. *J. Financ. Economet.*, 2017, **15**(3), 333–376.
Ang, A., Chen, J. and Xing, Y., Downside risk. *Rev. Financ. Stud.*, 2006, **19**(4), 1191–1239.
Artzner, P., Delbaen, F., Eber, J. M. and Heath, D., Coherent measures of risk. *Math. Finan.*, 1999, **9**(3), 203–228.
Bali, T. G., Comment on: Nonparametric tail risk, stock returns, and the macroeconomy. *J. Financ. Economet.*, 2017, **15**(3), 413–417.
Bali, T. G., Brown, S. J. and Caglayan, M. O., Macroeconomic risk and hedge fund returns. *J. Financ. Econ.*, 2014, **114**(1), 1–19.
Bali, T. G., Cakici, N. and Chabi-Yo, F., A generalized measure of riskiness. *Manage. Sci.*, 2011, **57**(8), 1406–1423.
Bali, T. G., Cakici, N. and Whitelaw, R. F., Maxing out: Stocks as lotteries and the cross-section of expected returns. *J. Financ. Econ.*, 2011, **99**(2), 427–446.
Bloom, N., The impact of uncertainty shocks. *Econometrica*, 2009, **77**, 623–685.
Bollerslev, T., Todorov, V. and Xu, L., Tail risk premia and return predictability. *J. Financ. Econ.*, 2015, **118**, 113–134.
Carhart, M. M., On persistence in mutual fund performance. *J. Financ.*, 1997, **52**(1), 57–82.
Dobrev, D. and Schaumburg, E., Comment on: Nonparametric tail risk, stock returns, and the macroeconomy (vol 15, pg 388, 2017). *J. Financ. Economet.*, 2017, **15**(3), 505–506.
Fama, E. F. and French, K. R., The cross-section of expected stock returns. *J. Financ.*, 1992, **47**(2), 427–465.

Fama, E. F. and French, K. R., Common risk factors in the returns on stocks and bonds. *J. Financ. Econ.*, 1993, **33**(1), 3–56.

Fama, E. F. and French, K. R., A five-factor asset pricing model. *J. Financ. Econ.*, 2015, **116**(1), 1–22.

Fletcher, R. and Leyffer, S., Solving mathematical programs with complementarity constraints as nonlinear programs. *Optim. Meth. Softw.*, 2004, **19**(1), 15–40.

Gul, F., A theory of disappointment aversion. *Econometrica*, 1991, **59**(3), 667–686.

Kelly, B. and Jiang, H., Tail risk and asset prices. *Rev. Financ. Stud.*, 2014, **27**(10), 2841–2871.

Kozak, S., Nagel, S. and Santosh, S., Interpreting factor models. *J. Financ.*, 2018, **73**(3), 1183–1223.

Krokhmal, P., Palmquist, J. and Uryasev, S., Portfolio optimization with conditional value-at-risk objective and constraints. *J. Risk*, 2001, **4**(2), 43–68.

Newey, W. and West, K., A simple, positive semi-definite, heteroskedasticity and autocorrelatoin consistent covariance matrix. *Econometrica*, 1987, **55**(3), 703–708.

Pstor, U. and Stambaugh, R., Liquidity risk and expected stock returns. *J. Polit. Econ.*, 2003, **111**(3), 642–685.

Rockafellar, R. T. and Uryasev, S., Optimization of conditional value-at-risk. *J. Risk*, 2000, **2**(3), 21–41.